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RESEARCH ARTICLE

Deep Learning Innovations for Underwater Waste Detection: An In-Depth Analysis

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ABSTRACT Underwater waste detection is a critical challenge for preserving aquatic ecosystems, particularly due to inherent underwater distortions such as light refraction, occlusion, and scattering. In this study, we present a novel deep learning framework for real-time underwater waste detection by evaluating state-of-the-art object detection algorithms on a manually annotated custom dataset comprising images across various water bodies to represent real-world turbidity, illumination, and occlusion. Our approach incorporates robust feature extraction and specialized data augmentation techniques that effectively mitigate the adverse effects of underwater distortions. We investigate multiple architectures, including YOLOv8n, YOLOv7, YOLOv6s, YOLOv5s, Faster R-CNN, and Mask R-CNN, analyzing key parameters such as mean average precision (mAP), inference speed, and computational efficiency. Our results demonstrate that YOLO-based models achieve faster & accurate performance, with YOLOv8n and YOLOv7 reaching a detection accuracy of 96%, compared to 81% and 83% for Faster R-CNN and Mask R-CNN, respectively. Computationally, YOLOv8n achieves 76 ms (milliseconds) per frame, while YOLOv5s runs at 14 ms per frame, confirming real-time viability for AUV deployment. The proposed approach offers significant advantages over existing methods by enabling rapid, accurate detection with low computational overhead, thereby paving the way for integration with autonomous underwater vehicles (AUVs) for environmental monitoring and waste management.

INDEX TERMS Computer vision, image processing, robotics, object detection, underwater trash, image segmentation, deep learning, YOLO, CNN.

I. INTRODUCTION

In recent years, underwater debris has become a significant environmental concern due to poor waste management, littering, and the expansion of global industries [1]. This growing accumulation of waste has led to multiple ecological problems, including the impairment of aquatic life and widespread water pollution [2], [3], [4], [5], [6]. Moreover, debris sinking into deeper epipelagic layers causes long-term environmental damage [7], [8], [9], polluting the water [10] and harming aquatic animals. The extraction of submerged debris is a challenging and cost-inefficient process, emphasizing the need for solutions that are both effective and resource-efficient. This raises a requirement for a solution

that can work efficiently in a variety of settings at low costs requiring minimal computational needs while delivering high inference speeds. In response to this, autonomous underwater vehicles (AUVs) offer a promising solution for debris detection and removal. With the latest advancements in the field of robotics and AI, [10], [11] it is possible to remove submerged debris with intelligent robots. However, current approaches are hindered by high costs, slow detection speeds, and significant computational and memory requirements, making them unsuitable for widespread deployment. Additionally, many existing datasets are limited to specific environments, reducing their applicability in diverse underwater settings and limiting the development of generalized models.

Despite these challenges, there is a lack of robust, real-time detection systems that are both accurate and

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computationally efficient across diverse underwater conditions. Variations in water clarity, lighting, turbidity, and other environmental factors further exacerbate the challenge of generalizing deep learning models to different underwater scenarios. This study addresses these gaps by evaluating several state-of-the-art deep learning architectures and proposing a framework optimized for integration with autonomous underwater vehicles (AUVs). To ensure broad generalizability, our contributions include the development of a diverse underwater waste dataset—constructed with varying environmental characteristics—and a comprehensive performance analysis that highlights the advantages of YOLO-based models in real-world environmental monitoring applications. The primary goal is to demonstrate the feasibility and efficiency of real-time visual detection of underwater debris, which is crucial for integrating detection systems into robotic platforms used for waste removal. This research aims to provide a baseline for object localization techniques, paving the way for more cost-effective, rapid, and scalable solutions in underwater trash detection.

The structure of this paper is as follows: review related work in the field of underwater debris detection, focusing on existing detection frameworks and challenges. Then introduces dataset modifications and the methodologies employed in our study, including the deep learning models and also present the experimental setup, followed by an analysis of the results. Finally discusses the implications of our findings and suggests potential avenues for future research.

Contributions:

1. We utilized our previously developed a diverse, publicly available custom underwater waste dataset which is an amalgamation of diverse 10k underwater images collected from open-source datasets [12], [13], [14], [15] and YouTube videos, manually annotated to contain waste with real-world turbidity, occlusion and lighting conditions [15] to conduct a comprehensive performance evaluation of state-of-the-art deep learning models—focusing on Naïve architectures and YOLO-based architectures—for real-time detection under challenging underwater conditions; and
2. Comprehensive benchmarking of six SOTA detectors against existing marine-debris datasets. We benchmarked the practical applicability of Naïve and YOLO based frameworks for integration with autonomous underwater vehicles (AUVs), by benchmarking metrics like Inference speed and providing a robust baseline for future research in underwater waste management.

II. REVIEW OF EXISTING LITERATURE

The growing crisis of underwater trash poses significant threats to marine ecosystems, requiring urgent attention and innovative solutions. This review explores the challenges

in detecting marine debris, advancements in technology, and the transformative role of deep learning in underwater debris detection. It also examines key datasets, environmental conservation efforts, alternative innovations, and the broader scope of detection models. Together, these insights pave the way for sustainable approaches to addressing marine pollution and its ecological impacts.

A. CHALLENGES IN UNDERWATER TRASH DETECTION

The detection and management of marine debris are critical to preserving marine ecosystems. The primary challenge lies in the debris' small size relative to the vast ocean, its submerged nature, and the potential for concealment on the sea bottom [16]. A comprehensive review has been done on the various existing literature in this domain explaining the problems faced in harsh marine environments, shown in Figure 1.

B. ADVANCEMENTS IN DETECTION TECHNOLOGIES

1) DEEP LEARNING AND OBJECT DETECTION

Advancements in deep learning have propelled the development of object detection algorithms, significantly enhancing marine debris detection capabilities. Figure 1 [17] shows a taxonomy of deep learning-based object identification, including detection components, learning algorithms, and applications and benchmarks. This comprehensive approach underpins the ongoing research into autonomous robotic systems [18], [19], [20] for environmental monitoring at both surface and underwater levels to innovative detection [3], [21], [22] and documentation [23] methods post-natural disasters using ultrasonic [13], [24], [25] and sonar [26] sensors in various aquatic environments [27], [28].

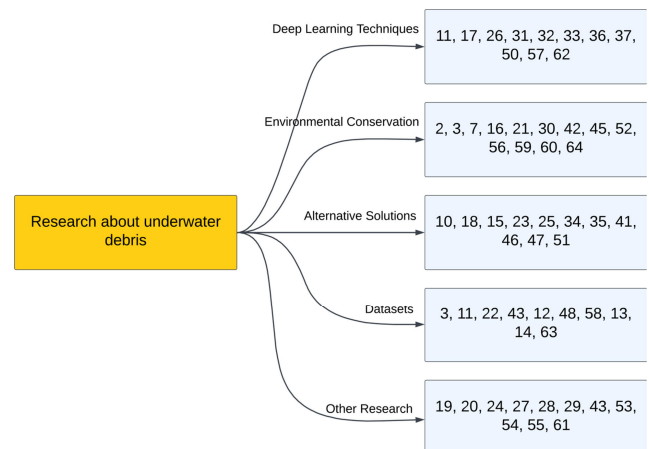


FIGURE 1. Literature available for the subsequent topics.

2) TECHNOLOGICAL INNOVATIONS IN DEBRIS DETECTION

Technological innovations extend beyond deep learning approaches that utilize neural networks [29], [30], [31], [32], [33] and image processing [34], [35], [36], incorporating LIDAR [25], sonar [26], and light scattering

techniques [27], [28] for underwater image enhancement. The operational use of multispectral images by autonomous aerial vehicles for macro-litter mapping and categorization represents another significant leap forward [20].

C. DEEP LEARNING'S PIVOTAL ROLE IN UNDERWATER DEBRIS DETECTION

Deep learning not only facilitates the detection of marine debris but also enhances the quality of underwater imagery. Notable studies employing deep learning for debris detection include the works of ocean pollution detection [29], identification of floating plastic [30], deep sea debris detection [31], [32], submerged debris detection [33], haze removal [34], localization of trash [35], as well as a hybrid loss function considering illumination and detection problems [37] and custom networks for detection [36]. These efforts demonstrate the efficacy of deep neural networks in identifying underwater debris, with innovations such as the 'YOLO- TrashCan' network emerging as a potent tool for marine debris identification [35], [36].

Recent studies have also explored advanced data augmentation techniques [38] and transformer-based architectures [39] to further enhance the robustness of underwater detection systems, thereby complementing the methods discussed above and emphasizing the importance of integrating uncertainty quantification in deep neural networks for environmental applications [40].

D. DATASETS AND ENVIRONMENTAL IMPACT ASSESSMENTS

The development of specialized datasets [30], [41], [42], [43], [44] like UIDEF [41], along with studies on the distribution of deep-sea litter [42] and beach litter segmentation [43], provides critical resources for training and evaluating detection algorithms. The contributions of research institutions, exemplified by the Monterey Bay Aquarium Research Institute's long-term data collection [44] and JAMSTEC's dataset compilation [22], are invaluable to the field. In addition, the TrashCan [12] Dataset has recently emerged as a valuable resource, further enriching the available datasets for training and evaluating detection algorithms.

E. INTEGRATING ENVIRONMENTAL CONSERVATION EFFORTS

Technological innovations are pivotal for environmental conservation, particularly in addressing marine debris and ecosystem degradation. Clear Blue Sea's Floating Robot for Eliminating Debris (FRED) initiative [3] deploys autonomous robots equipped with advanced sensors and navigation systems to detect and collect debris in coastal waters. FRED, designed to handle diverse marine environments, is engineered to navigate autonomously, allowing it to efficiently capture floating debris before it reaches the open ocean. The robot's design is modular, enabling it to be adapted for different debris sizes and ensur-

ing versatile functionality. Similarly, The Rozalia Project employs underwater rovers, notably the ROV (Remotely Operated Vehicle) Hector [7], which is specifically designed for oceanic cleanup operations. These rovers are equipped with specialized tools that enable the collection of debris from sensitive marine environments like coral reefs, minimizing disturbances to marine life. The Rozalia Project's multifaceted approach includes public education and scientific research, emphasizing the importance of maintaining marine ecosystems while exploring innovative solutions for debris removal.

Both initiatives illustrate the potential for robotics and technology to address marine pollution on a global scale, demonstrating the effectiveness of targeted interventions in preserving marine biodiversity and reducing ecosystem degradation. These projects highlight the transformative potential of technology in preserving marine ecosystems [25], [44], [45], [46], emphasizing the importance of technological innovation in combating marine debris and safeguarding the environment.

F. EXPLORING ALTERNATIVE SOLUTIONS AND INNOVATIONS

The exploration of alternative solutions, such as the NOAA's monitoring initiatives [45], which utilize satellite imagery and advanced data analytics to track and predict debris accumulation patterns, and Howell's investigation into marine detritus [44], which leverages field research and oceanographic modeling to understand the sources and distribution of debris, signifies a multifaceted approach in understanding and mitigating marine debris. These efforts aim to develop effective strategies for debris management and prevention by providing crucial data on debris hotspots and movement, thus guiding targeted cleanup operations and informing policy decisions.

These initiatives are instrumental in enriching datasets for underwater object detection algorithms by providing valuable real-world observations. NOAA's monitoring efforts offer comprehensive data on marine environments, aiding in algorithm refinement by enhancing the accuracy of detection systems. Howell's research delves into the composition and distribution of marine debris, offering crucial insights that contribute to algorithm optimization and the overall effectiveness of debris detection methods.

G. BROADENING THE SCOPE OF MARINE DEBRIS DETECTION MODELS

The broad scope of marine debris detection models is evident in the recent contributions to the field, with studies [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], [57], [58] emphasizing diverse approaches. Guo et al. [47] explore unsupervised underwater image enhancement using transformers, significantly improving image clarity for better detection. A comprehensive review of image analysis

methods for microorganism counting, including deep learning approaches, which are essential for understanding the impact of marine debris on microscopic life was performed by Li et al. [48]. Another comparative study on different learning algorithms to classify mangrove species using AAV multispectral images was performed by [49], aiding in the identification of marine debris in coastal ecosystems. Marin et al. [50] investigate deep-feature-based approaches to marine debris classification, revealing the potential of machine learning in accurately categorizing various debris types. Other researchers highlight the importance of accurate classification through a review and case study of marine microdebris [51] and also discuss the use of both remote and in situ devices for assessing marine contaminants [52], emphasizing the role of advanced technology in environmental monitoring. AAV imagery and deep learning are also utilized to assess anthropogenic marine debris in the Maldives [53], demonstrating the effectiveness of aerial surveys. Vashisht and Kumar [54] delves into object detection methods and image classification for marine debris, providing essential insights into identifying debris patterns [55]. Salgado-Hernanz et al. [56] underscore recent approaches in remote sensing for marine litter assessment, pointing toward future goals in debris detection. Mittal et al. in his research [57] focussed on underwater image classification through deep learning techniques.

Additionally, the classification of marine micro debris [51] and the use of remote sensing for marine debris detection [52], [53] further expand the methodologies available for marine conservation efforts.

H. CONTRIBUTIONS TO UNDERWATER IMAGE CLASSIFICATION AND DEBRIS DETECTION

Contributions to underwater image classification and debris detection are continually evolving, with significant overviews and studies provided by Vashisht and Kumar [54], [55], along with research in remote sensing [56], and survey-based study of deep learning architectures [57]. These contributions, along with the development of convolutional architectures for sonar images [58] and the Adaptive Lighting Enhancement algorithm [46], underscore the ongoing advancements in the field.

I. CONCLUDING REMARKS ON ENVIRONMENTAL IMPLICATIONS AND FUTURE DIRECTIONS

The detection and management of marine debris necessitate a comprehensive and multifaceted approach that incorporates deep learning, technological innovations, and concerted environmental conservation efforts. The body of research, including the operationalization of new detection models and the development of enriched datasets, lays a solid foundation for future advancements. Nonetheless, as the field progresses, it is crucial to consider the environmental implications of these technologies and strive for sustainable and ecologically responsible solutions.

III. DATASET

A. PUBLICALLY AVAILABLE DATASETS

Numerous open source underwater image datasets exist on the internet, but a common shortcoming is their high specificity and adaptation to the collection environment, introducing potential bias in the model. Some examples of these existing datasets include TrashCAN 1.0 [12] (both instance and material versions), Trash_ICRA19 [13], JAM-STEAC [14] and [15]. Current studies towards automatic waste detection using these datasets are hardly comparable due to the lack of benchmarks and widely accepted standards regarding the used metrics and data [59]. In our previous study [15], we introduced a custom dataset designed to address these challenges and enhance the efficiency of trash localization.

B. DATASET COMPARISON

A comprehensive comparison of all the properties like size, classes, collection location, water-type was performed on existing datasets [60] in order to identify each dataset's limitations and therefore utilizing this information to make the model trained on these datasets more robust. The diverse range of data prompted an analysis of these datasets which is available in the paper's source repository².

The used dataset [15] was constructed using three major steps: data collection, manual annotation, and data preprocessing using rotation, random cropping, horizontal flips, expansion, and color jittering [61]. This augmented approach to data has demonstrated notable enhancements in detection accuracy [15]. The final dataset contained 10,000 images of size 416×416 mapped into three classes: (i) Underwater Trash, (ii) Rover, and (iii) Biological life. We compare the different existing datasets relevant to this use case based on various factors, such as the number of classes, total images, environmental conditions, and the different classes, from Table 1.

IV. METHODOLOGY

With recent advancements in computer vision technologies and algorithms, this section introduces cutting-edge identification and categorization models, accompanied by established standards for the utilized dataset. The training metrics are then statistically evaluated for each model providing insights into their efficiencies and potential for real-world application.

A. DATASET DESCRIPTION AND PARAMETERS

The study utilizes a custom underwater waste dataset [15] comprising 10,000 images (416×416 pixels) categorized into three classes: Underwater Trash, Rover, and Biological Life. Data augmentation methods—including rotation, random cropping, horizontal flips, expansion, and color jittering—were applied to increase data diversity and enhance model robustness. The dataset is publicly

TABLE 1. Comparison of the classes present in various different datasets.

Dataset	No. of Classes	Bio	Paper and Glass	Metal waste and Plastic	Others	Unknown	Total Images	Environmental Conditions
Trash ICRA-19	3	1966	0	5051	0	0	6148	Normal Underwater
TrashCan 1.0	3	1009	0	2163	0	180	7212	Normal Underwater
TrashNet	6	0	1300	0	0	0	2527	Clear Background
Google search	1001	92	250	0	0	366	1320	Varying
Extended TACO	28	70	1190	6060	2800	150	1500	Wild Waste
Drinking waste	4	0	1160	3600	0	0	9640	Clear Conditions, Cans & Bottles
Classify-waste	3	160	3900	9660	2800	520	27500	Varying

available at [https://github.com/karanwxliaa/Underwater-Trash-Detection] for reproducibility.

B. OBJECT LOCALIZATION

The object detection architectures were carefully selected from the latest, highly coherent, and promising set of networks currently available. Each architecture has its own unique strengths and weaknesses, offering varying degrees of reliability, execution pace, and other metrics which we identify in this literature. In this paper we also leverage several neural network architectures considered to be state-of-the-art models, including YOLOv9, YOLOv8 nano, YOLO v7, YOLO v6s, YOLO v5s, and YOLO v4 Darknet, utilizing their respective source repositories [62]. In addition, custom training was performed on region-based Mask-RCNN and Faster RCNN architectures with Inception v2 [63].

C. OBJECT SEGMENTATION

Building on the advancements in computer vision, this section focuses on cutting-edge object segmentation methodologies. Real-time instance segmentation, exemplified by YOLACT and Mask R-CNN models trained on underwater litter data, offers promising prospects for litter detection, with Mask R-CNN achieving a mean average precision (mAP) of 0.377 and YOLACT providing faster detection speeds [64]. AquaSAM [65] enhances underwater segmentation, particularly in tasks like coral reef segmentation, showcasing a notable 7.13% improvement in Dice Similarity Coefficient. A novel deep learning-based method for trash trap defect detection [66] demonstrates significant enhancements in reliability, with error rates below 15% and accuracy rates exceeding 85%. Furthermore, leveraging Cycle Consistent Generative Adversarial Networks (GANs), enhancing unpaired underwater images exhibits superior performance over traditional techniques [67], showing potential for

various marine applications such as trash identification and coral reef inspection. State-of-the-art segmentation models are considered, aligning with the latest advancements in technology. The chosen models adhere to predefined standards for the dataset, ensuring a rigorous evaluation of the segmentation performance. Each segmentation architecture brings its distinctive features, strengths, and weaknesses, contributing to a comprehensive understanding of their efficacy.

D. GRAPHICS PROCESSING UNIT

In this study, we utilized a Tesla T4 GPU with a total memory capacity of 16GB, comprising 40 Streaming Multiprocessors (SMs) and a shared 6MB L2 cache across all SMs. Released in 2018, this GPU is equipped with 2560 CUDA cores.

E. ARCHITECTURES

This section provides an overview of latest state-of-the-art models employed in the research and presents the outcomes achieved during experimentation. Presently, object detection frameworks rooted in deep learning can primarily be categorized into two main groups: (i) two-stage detectors, exemplified by region-based CNN (R-CNN) [68] and its variations [69], [70], [71], and (ii) one-stage detectors, such as YOLO [72] and its variations [73], [74]. Two-stage detectors initially utilize a proposal generator to produce a sparse set of proposals and then extract features from each proposal. This is followed by region classifiers that predict the category of the proposed region. In contrast, one-stage detectors directly predict the category of objects at each location on the feature maps without the additional region classification step. While two-stage detectors commonly achieve superior detection performance and report state-of-the-art results on public benchmarks, one-stage detectors are notably more

time-efficient and exhibit greater suitability for real-time object detection [17].

1) YOU ONLY LOOK ONCE (YOLO)

YOLO, an acronym for You Only Look Once, stands out as a prominent and sought-after object identification method. It excels in the real-time identification of multiple items within a coexisting image or video. Unlike traditional methods, YOLO employs a single neural network to predict bounding boxes and class probabilities directly from the entire image in a single analysis. This approach enhances efficiency and precision, making YOLO a superior choice compared to other existing object detection and identification systems. A comparative analysis in Table 2 underscores YOLO's prowess, showcasing its superior performance in terms of model size, mean average precision (mAP), speed, and computational parameters across different variants like YOLOv8n, YOLOv8x, and YOLOv5s.

TABLE 2. Comparison between the computation values and speed of the best performing models.

Model	Size (pixels)	mAP val 50-95	Speed CPU ONNX (ms)	Speed A100 Ten-sorRT (ms)	params (M)	FLOPs (B)
YOLOv8n	640	37.3	80.4	0.99	3.2	8.7
YOLOv8x	640	53.9	479.1	3.53	68.2	257.8
YOLOv5s	640	56.8	98	6.4	7.2	16.5

- **YOLOv8x [75] (XLarge):** Building upon the innovations of its pre-decessors, YOLOv8x represents an advanced variant of YOLOv8, with a primary focus on achieving heightened accuracy, particularly for small and intricate objects. This improvement is realized through an augmentation in model size, achieved by enhancing both depth and width. The increased dimensions enable the model to capture more intricate visual patterns, resulting in superior detection performance, especially in challenging scenarios. YOLOv8x outperforms YOLOv8n, making it particularly well suited for applications demanding exceptional precision. Its larger model size enhances object understanding and localization, thereby significantly improving overall object detection capabilities.
- **YOLOv7 [76] (tiny/W6):** The YOLOv7 algorithm has markedly outperformed its earlier versions in terms of both speed and precision. Notably, it requires less sophisticated hardware compared to standard conventional architectures. YOLOv7 demonstrates an impressive ability to be trained effectively on smaller datasets without pre-learned weights within a shorter timeframe. However, it is important to highlight that,

despite these strengths, YOLOv7 falls short of achieving superior accuracy and inference speeds when compared to the latest YOLOv8 models.

- **YOLOv5s [77] and YOLOv6s [78] (Small):** They exhibit relatively consistent performances and results, making them suitable for benchmarking. However, they fall short when compared to some of the most widely used stable models for object identification and segmentation.

2) R-CNN

R-CNN [68], introduced by Girshick et al. in 2014, is a ground breaking two-stage object detection system. With respect to earlier novel methods like SegDPM [79], which achieved a mean Average Precision (mAP) of 40.4 % on Pascal VOC2010, R-CNN showed a significant improvement with 53.7% mAP. The R-CNN pipeline is divided into three main components: proposal generation, feature extraction, and the region classification.

R-CNN generates a sparse collection of proposals (about 2,000 for each image) using Selective Search [80]. The goal of this technique is to remove areas that are readily recognized as background. Next, before encoding each proposal into a feature vector, a deep convolutional neural network crops and resizes it into a fixed-size region (for example, 4,096 dimensions). Following that, a one-vs-all SVM classifier is used for region classification. Finally, using the retrieved characteristics as input, bounding box regression models are trained.

Inspired by spatial pyramid matching (SPM) [81], He et al. proposed SPP-net [82] to boost R-CNN's speed and learn more distinct features. Unlike the traditional method of clipping proposed regions and passing them separately into a CNN model, SPP-net uses a deep convolutional network to generate the feature map from the entire image. Next, using the Spatial Pyramid Pooling layer (SPP), it collects fixed-length feature vectors from the feature map.

3) FAST R-CNN

Fast R-CNN [69] is a multi-task learning detector that outperforms SPP-net while resolving its shortcomings. Similar to SPP-Net, Fast R-CNN generates a feature map for the full picture and extracts fixed-length region features from it. However, unlike SPP-net, Fast R-CNN uses a ROI (Region of Interest) Pooling layer to retrieve region features.

4) FASTER R-CNN

Even with great progress made in learning detectors, the proposal generation stage was still dependent on traditional methods like Selective Search [80] or Edge Boxes [83], which relied on low-level visual signals and could not learn based on data. To tackle this limitation, faster R-CNN, an improvement over R-CNN [68], which utilized a Region Proposal Network (RPN) [84], [85], [86] was introduced, enhancing the creation of an end-to-end trainable object detection network [87].

The RPN generates region proposals using the last convolutional feature map, followed by fully connected layers for final detection. Although many implementations typically employ VGG-16 as a standard for feature extraction, we chose to use Inception v2, which has demonstrated superior object identification performance in conventional datasets [88], [89]. However, it may misidentify patches in the background as objects due to its limited ability to consider large contexts.

When comparing generalization errors, YOLO algorithms surpass R-CNN by producing lower generalization errors. In comparison to the YOLO algorithms, Faster R-CNN exhibits slightly fewer localization errors but at the expense of higher computational needs and slower speed.

5) MASK RCNN

In order to enhance the overall flexibility of the process, He et al. [90] introduced Mask R-CNN, a model capable of predicting bounding boxes and segmentation masks at the same time, yielding optimal results. Huang et al. [91] developed Mask Scoring R-CNN, a system that prioritizes mask quality awareness. This framework learns the quality of predicted masks and uses calibration approaches to correct the misalignment between mask qualities and mask confidence score. This has also emerged as a faster variant by combining ResNet with a Feature Pyramid Network (FPN) the combination of which creates a feature extraction network capable of generating a multi-layer convolutional feature map from input images [78]. The Region Proposal Network (RPN) is then employed to generate object region proposals, while the detection head and mask head are utilized for object detection and instance segmentation. The algorithm is structured into two main components: detection and instance segmentation [90], [91].

Despite its effectiveness, Mask R-CNN can face challenges in complex maritime environments. External variables such as distracting objects and low image resolution may impact the algorithm's performance by making it difficult to extract object features, leading to reduced attention toward objects during detection.

F. EVALUATION METRICS

After model training, we utilize distinct evaluation and validation datasets, separate from the training dataset, to assess the accuracy of the architecture. The model accurately classifies trash by creating bounding boxes with a confidence score of 0.50 or higher. The amount of true positive bounding boxes containing marine plastic debris and the accuracy with which true negatives are detected are used to evaluate the model's performance.

These performance metrics were used to evaluate each model's performance, pros and cons:

- **True positive and True negative values:** These values are calculated by counting the number of bounding boxes correctly predicted as trash (true positives) and

the number of bounding boxes correctly predicted as not trash (true negatives).

- **Precision and Recall [92]:** Evaluates if the model foreseen trash located in the input image in which precision reflects the model's ability to correctly predict plastic waste, while recall reflects the model's ability to identify all instances of plastic waste in the input image. Precision is calculated as the ratio of true positives to the sum of true positives and false positives, while recall is calculated as the ratio of true positives to the sum of true positives and false negatives [92].

$$Recall = \frac{TP}{TP + FN} \quad (1)$$

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

These equations, 1 for recall and 2 for precision illustrate how the performance of the model is quantitatively determined in terms of its ability to identify plastic waste accurately.

- **Mean Average Precision [93], [94]:** This metric measures the accuracy of the model in identifying underwater trash waste across a set of input images. The mAP is calculated by building a precision-recall curve using the Intersection over Union (IoU) formula to compare the predicted and ground truth bounding boxes [93] as given Eq (3). The precision-recall curve is then integrated to obtain the mAP. After collecting the accurate and inaccurate data, we utilize the "Intersection over Union (IoU)" formula to give rise to a precision-recall curve:

$$IOU = \frac{BBox_{pred} \cap BBox_{GroundTruth}}{BBox_{pred} \cup BBox_{GroundTruth}} \quad (3)$$

where $BBox_{pred}$ and $BBox_{GroundTruth}$ represent the expected areas under the curve for ground truth and predicted bounding boxes, respectively. In order to maximize accuracy, it is necessary to set a high threshold for confidence and IoU, ensuring that correct predictions exceed this threshold. Subsequently, the average precision (mAP) can be calculated by integrating the precision-recall curve [94] as given in Eq(4):

$$mAP = \int_0^1 p(x) dx \quad (4)$$

- **Processing speeds:** This metric denotes the average time taken by the model to perform pre-processing, inference and post-process for the detection and identification of underwater trash in an input image, measured in milliseconds per image.
- 1) **YOLOv8n** gave us an average speed of 1.9ms for pre-process, 73.5ms for inference and 0.95ms post-process per image at shape (1, 3, 416, 416)
 - 2) **YOLOv8x** gave us an average speed of 0.3ms for pre-process, 16.6ms for inference and 1.4ms post-process per image at shape (1, 3, 416, 416)

- 3) **YOLOv5s** gave us an average speed of 0.5ms for pre-process, 12.0ms for inference with 1.2ms post-process per image at shape (1, 3, 640, 640)

V. RESULTS

1) EVALUATING RESULTS

A comprehensive analysis of individual components for the best performing frameworks was conducted, involving thorough examinations of trash data publicly available in various scenarios, including clean water, natural and man-made water bodies such as lakes and ponds, surroundings, and ocean beds was performed. The utilized architectures also demonstrate a relatively high F1 score and a high mean-average precision (mAP) score relative to their inference speeds. Table 3 shows the results of an in-depth evaluation comparing multiple architectural networks. The observed trade-offs indicate that the outcomes derived in this research provide a more accurate representation of performance over time across a broader spectrum of water conditions, allowing for an in-depth analysis of the various model's object localization performance in this domain.

From Table 3 we conclude that both YOLOv8-nano and YOLOv8-XLarge successfully achieve real-time localization of epipelagic detritus, demonstrating robust metrics. Figure 2 shows the precision and recall values for YOLOv8n and YOLOv8-XLarge.

Additionally, advanced algorithms within the same family, including YOLOv5, v7, YOLO-NAS, and YOLO-SAM, were also evaluated, yielding similar outcomes. The decision was made to opt for the lighter models like tiny, small and nano networks due to their quickest evaluation performance, fewer parameters, and lower processing power requirements. These algorithms also outperformed classic Mask R-CNN and Faster R-CNN algorithms, showcasing significantly higher speeds & better accuracy. The model's performance in real-world deployments mirrored the results obtained in controlled evaluations, with only minor deviations between the outcomes presented in Table 3 and those observed in a real-time environment. Figure 3 shows Confusion matrix for the best performing YOLOv8 model.

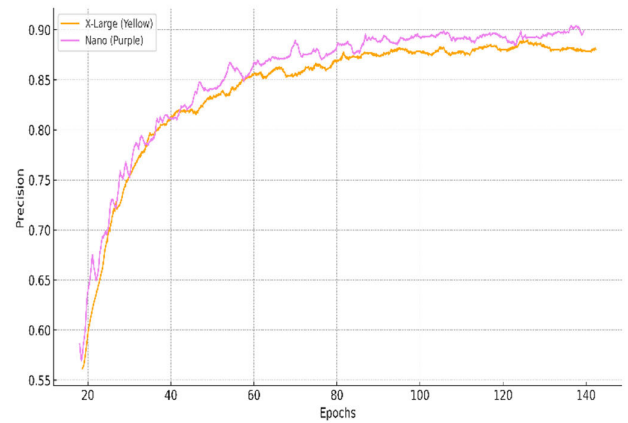
2) VISUALIZING RESULTS

As part of its image processing, the network captures two important data points in arrays: Scores, which reflect the degree of certainty for the predicted boxes, and the

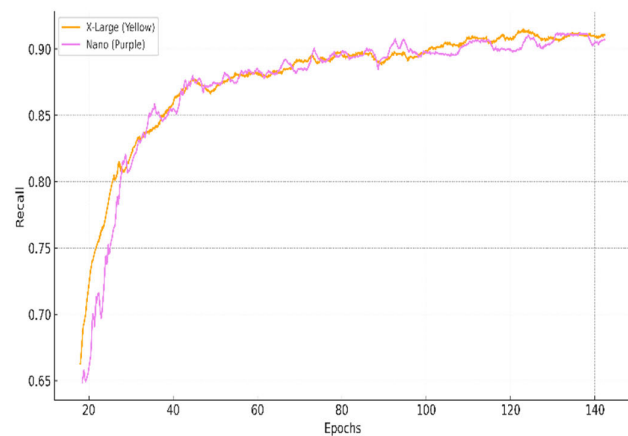
total number of Detections made on the image. In order to render the bounding boxes on the images, the network converts the normalized coordinates to image coordinates, utilizing the Equation 5.

$$ImgCord_k = Boxscore_i^j * Width \quad (5)$$

In the above equation, k represents all corners (right, left, top, & down), j belongs to the set 0, 1, 2, 3, i denotes the box index, and Width indicates the image width. During the evaluation of test images, the coordinates of the image were utilized to visualize the obtained results of the



a.) Precision



b.) Recall

FIGURE 2. a) Precision and (b) Recall values for the suggested algorithms YOLOv8n and YOLOv8x with colors purple and yellow respectively.

TABLE 3. Comparison between performance metrics of the models.

Network	mAP	Preci on	Reca ll	Box- Loss	Cls- Loss	Obj- Loss	Epoch
YOLOv8 n	0.96	0.94	0.92	0.02 2	0.00 05	0.01 1	170
YOLOv7	0.96	0.96	0.93	0.01 9	0.00 05	0.00 8	120
YOLOv6 s	0.90	0.94	0.92	0.08	0.07	0.45	110
YOLOv5 s	0.96	0.95	0.93	0.02	0.00	0.00	180
Faster R- CNN	0.81	0.88	0.85	0.08	0.03	0.02	100
Mask R- CNN	0.83	0.85	0.87	0.09	0.01	0.08	100

bounding boxes. The test dataset, which was deliberately designed to provide a representation of unseen test data new to the model to test the model's performance on new data

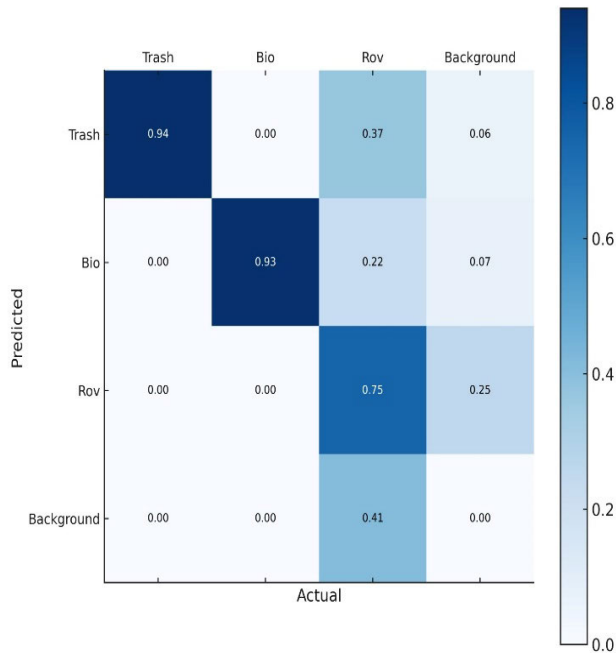


FIGURE 3. Confusion matrix for the best performing YOLOv8 model.

was used to obtain the results presented in Figures 4,5 and 6. Details on the dataset [15] benchmarks can be found in the Dataset Section III.

In addition to analyzing the models using our test set, we also subjected each network to two additional videos, as illustrated in the Figure 6 One of the videos shown in Figure was a viral footage of ocean waste captured in Bali [95], which presented a unique and previously unseen visual context compared to our training set. Despite the significant challenges posed by this video to all our detectors, these models were able to provide promising detection.

Figure 5 depicts real-time segmentation results utilizing the YOLOv8 [79] which can further be used for quantization of trash in a particular environment.

3) COMPARATIVE ANALYSIS AND DISCUSSION

Our experiments demonstrate that one-stage YOLO detectors (YOLOv8n and YOLOv7) not only achieve state-of-the-art accuracy but also deliver the low latency essential for on-board AUV operation. Both YOLO variants reach a mAP@0.5 of 96% a 15-point gain over Faster R-CNN (81%) and a 13-point gain over Mask R-CNN (83%) while maintaining end-to-end frame times under 80 ms on a Tesla T4 GPU. This speed–accuracy balance makes YOLO models uniquely suited to real-time underwater waste detection: they can process HD video streams at >12 fps, enabling continuous monitoring without prohibitive power or compute requirements. Beyond aggregate mAP, we observed class-specific strengths and weaknesses. YOLOv7 slightly outperforms YOLOv8n on recall (0.93 vs. 0.92) for small “Trash” objects



a) Batch results for Faster-RCNN [64]



b) Batch results for YOLO [73] models

FIGURE 4. Batch comparison between the detection outputs of a) RCNN & b) YOLO models.

in clear water, whereas YOLOv8n exhibits more stable precision under high turbidity, likely due to its newer backbone’s improved feature fusion. In contrast, two-stage detectors suffer from slower region-proposal overheads and are more

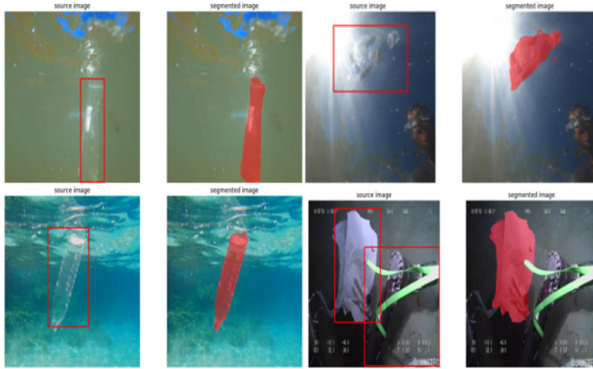


FIGURE 5. Real-time segmentation results for YOLOv8 models.

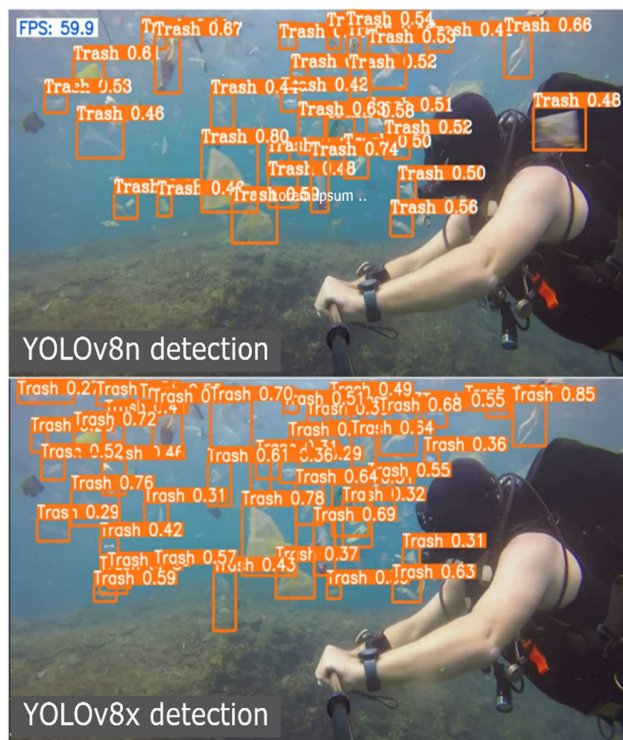


FIGURE 6. Performance on new, unseen test data using the best tradeoff model [70].

prone to missed detections when debris is partially occluded or sits against complex biological backgrounds. Despite these gains, all models experience a modest 4–6 % mAP drop in low-light or highly turbid conditions, highlighting the need for targeted augmentation (e.g., simulated turbidity) or uncertainty-aware postprocessing for safety-critical AUV missions. Future work will explore lightweight attention modules, domain-adaptive fine-tuning, and real-time mask refinement to further bolster robustness in the most challenging underwater environments. Overall, our findings confirm that lightweight YOLO architectures offer the best trade-off for embedded, real-time AUV deployments in marine waste management.

VI. CONCLUSION

This work evaluated both naïve and YOLO-based computer vision architectures for detecting and quantifying submerged marine debris across epipelagic, mesopelagic, and local water bodies. It demonstrated that YOLOv5, YOLOv7, and YOLOv8 models achieved up to 96 % mAP with sub-80 ms inference times, outperforming Faster R-CNN and Mask R-CNN in both speed and accuracy. We also developed a 9.6 k-image underwater waste dataset and delivered a comprehensive benchmark of multiple state-of-the-art detectors. Despite these advances, limited dataset diversity and reduced performance in low-light conditions remained as challenges. Future work will expand the dataset to encompass more varied underwater environments, incorporate uncertainty quantification for safer AUV deployments, and explore transformer-based enhancements to improve robustness under challenging visual conditions. We further plan to deploy and validate the optimized detection framework on autonomous underwater vehicles for continuous marine debris monitoring and rapid response.

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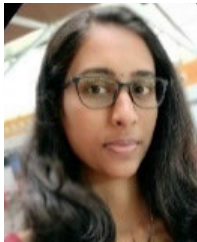
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